

START

LOAD households from "households.txt" into a list

LOAD floods from "floods.txt" into a list

IF files do not exist, start with empty lists

REPEAT

SHOW menu (1 Register, 2 Check readiness, 3 Log flood,
4 Street summary, 5 Search, 6 Exit)

ASK user for choice

IF choice = 1:

ASK name, street, member count, vulnerable counts

VALIDATE name not blank, members ≥ 0 , vulnerable $\leq$ members

IF invalid: SHOW error, back to menu

GENERATE household id

ADD household to list

SAVE list to file

SHOW "Registered with id <id>"

ELSE IF choice = 2:

ASK household name or id

IF not found: SHOW "No matching household", back to menu

ASK rainfall (light/moderate/heavy)

ASK river level (normal/alert/alarm/critical)

VALIDATE both values are accepted

readiness = 1

IF river = "critical": readiness = 4

ELSE IF river = "alarm":

IF rainfall = "heavy": readiness = 4 ELSE readiness = 3

ELSE IF river = "alert":

IF rainfall = "heavy": readiness = 3 ELSE readiness = 2

ELSE IF rainfall = "heavy": readiness = 2

LOOK UP action text and base checklist for readiness

IF household has children: ADD child items

IF household has elderly: ADD medicine/mobility items

IF household has PWD: ADD assistance items

SHOW readiness, reason, action, household summary, checklist

ELSE IF choice = 3:

ASK household name or id

IF not found: SHOW "No matching household", back to menu

ASK flood date, depth (cm), duration (hrs)

VALIDATE depth ≥ 0 and duration ≥ 0
ADD flood record to log
SAVE log to file
SHOW "Flood event recorded"

ELSE IF choice = 4:
count flood events per street
SORT streets by count, highest first
SHOW each street with count and most recent date

ELSE IF choice = 5:
ASK name or street to search
COLLECT households whose name or street contains the text
IF none: SHOW "No matches found"
ELSE SHOW each match with id and street

ELSE IF choice = 6:
SHOW "Data saved. Exiting."
STOP

ELSE:
SHOW "Invalid choice. Enter 1 to 6."
END REPEAT
END